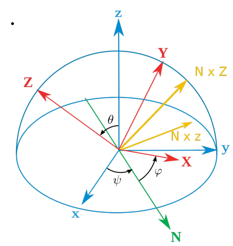
Homework №4

Deadline: *Wednesday, June 24, 2026*

1. (***2 points***) Using hamiltonian formalism, analyze the motion of the axially symmetric rigid body ("symmetric top") in the homogeneous gravitational field. Assume that the orientation of the rigid body is parametrized in terms of the Euler angles α, β, γ in ZXZ convention, so the components of the angular velocity Ω are given by



$$\Omega_1 = \dot{\alpha}\sin\beta\sin\gamma + \dot{\beta}\cos\gamma,\ \Omega_2 = \dot{\alpha}\sin\beta\cos\gamma - \dot{\beta}\sin\gamma,\ \Omega_3 = \dot{\alpha}\cos\beta + \dot{\gamma}$$

in the body's rest frame (can get this result making geometric projections).

- **(1/2 point)** Construct the lagrangian and the hamiltonian for this system
- **(1/2 point)** Write out the canonical equations of motion. Identify the cyclic variables and the integrals of motion.
- **(1 point)** Write out the Hamilton-Jacobi equation for this system and integrate it.

  Note: *The goal of this part is to test your knowledge of the Hamilton-Jacobi formalism, for this reason solution with any other method will not be taken into account.*

  Help: *The integration of the equations of motion and many intermediate expressions may be simplified using the variable* $\zeta = \cos\beta$ *instead of* β.

2.(***2 points***) A relativistic pointlike mass m moves in the Coulomb field $U = -\kappa/r$. Assume now that the particle moves slowly ($v \ll c$), so the difference between the relativistic and nonrelativistic hamiltonians can be considered as a a small perturbation,

$$\Delta H = H - H_0 = \sqrt{p^2c^2 + m^2c^4} - \left(mc^2 + \frac{\boldsymbol{p}^2}{2m}\right) \approx -\frac{\left(\boldsymbol{p}^2\right)^2}{8\left(c^2m^3\right)}$$

$$H_0 = mc^2 + \frac{\boldsymbol{p}^2}{2m} + U(r)$$

where $\boldsymbol{p}$ is the momentum of the particle.

- **(1 point)** Using canonical perturbation theory, find explicitly the first correction to the trajectory of the particle. Find the precession of the periapsis (closest point) due to this correction.
- **(1 point)** In Lecture 15 we solved the relativistic Kepler problem exactly, and afterwards made expansion in the final solution in the $c \to \infty$ limit. Analyze if the two methods of solution give equivalent results for the first correction.

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3. (**2 points**) A pointlike mass m moves in the field of time-dependent spherical harmonic oscillator

$$U(r) = -\frac{k(t)r^2}{2}$$

where the parameter k increases very slowly (*adiabatically*) as a function of time. Analyze how the trajectory of the mass will change as a function of time.

- (**1 point**) Identify the adiabatic invariants for this system and express them in terms of the parameters which characterize the orbit of the particle (its size, shape and period of motion).
- (**1 point**) Analyze how the energy E and angular momentum M of the system depend on time.